February 23, 2026

TO: U.S. Department of Health and Human Services
Re: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care (RIN 0955-AA13)

On behalf of the American Association of Clinical Endocrinology (AACE), thank you for the opportunity to respond to this RFI. We support responsible, patient-centered adoption of AI in clinical care that strengthens trust, protects safety and privacy, and improves real-world outcomes while reducing clinician burden.

This response was developed by Dr. S. Sethu K. Reddy, MD, MBA and David Lieb, MD on behalf of the AACE Advocacy Committee. For questions, please contact Ms. Surea Gainey, Advocacy & Regional Engagement Manager, at sgainey@aace.com.

Sincerely,

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Elevating the practice of clinical endocrinology to improve global health

Introduction

The American Association of Clinical Endocrinology (AACE), founded in 1991, is a global, inclusive community of more than 5,700 endocrine-focused clinical members, affiliates, and partners. Our mission is elevating the practice of clinical endocrinology to improve global endocrine health, and our vision is achieving healthier communities through endocrine innovation, education, and care. AACE members provide care for people living with endocrine and metabolic disorders across the United States, working as part of multidisciplinary endocrine care teams. Endocrinologists have been at the forefront of technology-assisted delivery of healthcare for decades. We appreciate the opportunity to respond to this request for information because artificial intelligence (AI) will increasingly influence how endocrine care is delivered. Recent clinical experience suggests AI can reduce documentation burden and improve detection and outcomes in some settings, but real-world benefit varies based on workflow integration and implementation quality, reinforcing the need for rigorous evaluation and oversight [1]. Patient safety and data protection are paramount. HHS policies should also promote improved clinical outcomes, clinician well-being, and increased access for all to technological advances allowing acceptable cost/benefit ratios.

To make this RFI actionable, we urge HHS to focus on a few practical steps. First, adopt a risk-based approach for clinical AI that clarifies expectations for transparency, validation, and how performance should be monitored and updated after deployment. Second, align payment and quality programs so tools that measurably improve outcomes and reduce clinician burden are supported. Third, accelerate interoperability by advancing standardized data exchange (FHIR/USCDI and related vocabularies) and by requiring clear provenance when AI-generated content enters the medical record. Fourth, publish operational guidance for governance, privacy, and cybersecurity consistent with HIPAA and other applicable federal and state laws, including auditability, vendor accountability, and incident response expectations. Finally, invest in pragmatic research and real-world demonstrations—especially in community, rural, and under-resourced settings—so benefits don't remain limited to a small number of well-resourced health systems. Unlike the typical static approval of clinical technology, AI-based technology evolves rapidly and thus requires a parallel regulatory governance system to be both flexible and adaptable.

Question 1: What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

Private-sector innovation and sustained clinical use of AI are most constrained by the difficulty of building tools that are safe, generalizable, and easy to use in real clinical workflows. Data needed for development and validation remain fragmented across electronic health records, claims systems, and devices, and are often labeled inconsistently or missing key context (for example, medication timing and discontinuations, dose changes after subspecialty consultation, and whether data reflect inpatient versus outpatient care). This slows development, increases patient-safety risk when tools are moved across sites, and makes it harder to ensure equitable performance for underrepresented and lower socioeconomic groups [2].

Adoption also stalls when AI increases clicks, alerts, or documentation rather than reducing work. It also stalls when model outputs are not transparent enough for clinicians to trust and appropriately oversee them, and when clinicians have not been trained on appropriate use and limitations. In addition, vendors and health systems face ongoing costs for integration, cybersecurity, and monitoring for performance drift and safety events, with uncertain reimbursement and unresolved liability/accountability for AI-supported decisions. Finally, the "pathway" to commercialization is less predictable than in other industries because requirements span multiple domains (privacy, interoperability, reimbursement, and—in some cases—medical device regulation), and expectations for validating and maintaining adaptive models remain unclear. In endocrinology, these barriers are amplified by high-stakes medication decisions (insulin and glucocorticoids), longitudinal diabetes technology data (CGM and insulin delivery), and the need to coordinate issues such as thyroid imaging, pathology, and follow-up across settings [2-4].

Question 2: What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

HHS can accelerate safe adoption by reducing integration burden, clarifying expectations for lifecycle oversight, and aligning payment incentives with measurable improvements in patient safety and outcomes. A high-impact step is strengthening interoperability expectations through ONC certification

(45 CFR Part 170) and information-blocking enforcement (45 CFR Part 171) so core clinical data and relevant device data can be exchanged through standardized, secure application programming interfaces (APIs) rather than custom interfaces. This is particularly important for endocrine care, where continuous glucose monitoring (CGM) and insulin delivery data influence day-to-day decisions and inpatient safety, and where poor data flow can increase clinician workload and worsen inequities in access [5, 6]. HHS should also coordinate with FDA to support a lifecycle approach for higher-risk clinical AI, including pre-authorized update pathways for adaptive tools (e.g., predetermined change control plans), quality management expectations (21 CFR Part 820), and postmarket safety surveillance and reporting (21 CFR Parts 803 and 822) so performance drift, bias, and safety events are detected and addressed over time [5].

HHS should enable privacy-preserving multi-site evaluation while maintaining strong patient protections by providing clearer, more operational guidance under HIPAA Privacy and Security Rules (45 CFR Parts 160 and 164) that supports responsible use of limited datasets and privacy-enhancing approaches (e.g., federated or distributed analytics) for validation and monitoring, paired with clearer national expectations for de-identification and data provenance for AI development [5, 6]. On the payment side, CMS should create predictable pathways that reward demonstrated reductions in preventable harm and administrative burden (for example, inpatient hypoglycemia prevention and streamlined prior authorization documentation for obesity and diabetes therapies), while discouraging incentives that drive unnecessary or duplicative testing or other utilization without benefit; relevant levers include Medicare coverage and payment policies (42 CFR Parts 410 and 414) and hospital payment innovation pathways (42 CFR Part 412), including time-limited transitional coverage while real-world outcomes and economic value are evaluated [5, 6]. This 'high-value AI' paradox—where accurate tools can still increase downstream utilization and costs without improving outcomes unless incentives and workflows are aligned—reinforces the importance of payment and quality program alignment for AI adoption [7].

Finally, HHS should revisit program integrity rules to permit responsible deployment partnerships through targeted updates or clarifications to Stark and Anti-Kickback frameworks (42 CFR Part 411 and 42 CFR Part 1001) when arrangements are transparent, not tied to referral volume, and demonstrably improve safety, quality, and equitable access; these safeguards are especially important given documented risks of hidden bias in deployed models [8]. Any clarification should be narrowly tailored to bona fide value-based arrangements, conditioned on documented quality and patient-safety outcomes, fair-market-value protections, and transparent auditing and reporting to prevent misuse. Programmatic investments that support rural implementation (connectivity, telehealth, and remote monitoring) can further ensure AI-enabled benefits reach underserved communities rather than widening disparities.

Question 3: For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (*e.g.,* relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

Non-medical-device AI (for example, administrative copilots, documentation tools, scheduling/triage systems, consumer apps and wearables, and patient-facing chat tools) can still influence clinical decisions and therefore patient safety, even when marketed as "wellness" or "administrative" tools and not reviewed as medical devices. Novel issues include unclear allocation of liability when an AI-generated summary, triage suggestion, or coaching recommendation contributes to harm; vendor indemnification and contracting practices; privacy risks and secondary use of health-related data (including use for model training); and cybersecurity risks from new interfaces and third-party integrations. These gaps are especially relevant in endocrinology because patient-facing tools may affect insulin dosing decisions, hypoglycemia responses, or weight-loss medication use, and errors can cause immediate harm [9].

HHS can help by promoting a risk-based governance approach tied to function and impact, not marketing claims: clear disclosure when AI is used; minimum privacy and security expectations (including clearer expectations when HIPAA does and does not apply); documentation of intended use, limitations, and appropriate clinician oversight for higher-risk uses; and standardized pathways for reporting and learning from AI-related near-misses and adverse events. HHS guidance can also promote transparency and subgroup performance monitoring, and require accessibility considerations (language, disability, and rural broadband constraints) so adoption improves outcomes without widening disparities. These steps are consistent with emerging responsible-use guidance and the growing literature calling for clearer accountability standards for clinical AI [10, 11].

Question 4: For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

Evaluation should be matched to clinical risk and should measure real patient outcomes and workflow impact, not only model accuracy. Before deployment, promising approaches include independent external validation on datasets from multiple health systems (including community settings and underrepresented groups); assessment of calibration (whether predicted risks match observed outcomes); robustness testing across sites, devices, and documentation patterns; and human-centered workflow testing to confirm the tool reduces cognitive load and documentation burden rather than creating alert fatigue or burnout. Pre-deployment review should also include privacy and cybersecurity evaluation for third-party integrations and clear documentation of intended use, limitations, and appropriate oversight. In endocrinology, examples include inpatient hypoglycemia risk prediction, diabetes technology decision support using CGM data, and thyroid ultrasound decision support—each of which can behave differently across patient populations, equipment, and care settings, and requires appropriate clinician training and oversight to ensure accurate interpretation [3, 11, 12].

After deployment, ongoing monitoring is essential: drift and calibration checks, routine performance reporting by subgroup (race/ethnicity, language, socioeconomic proxies, rurality), and a standardized incident reporting process when AI contributes to near-misses or harm. HHS support would be impactful through funding pragmatic trials and implementation science, supporting shared evaluation datasets and toolkits, and enabling safety-learning infrastructure and governance models (e.g., multidisciplinary oversight committees) that help organizations respond quickly when performance degrades or inequities emerge. Mechanisms such as contracts and cooperative agreements could build common evaluation infrastructure, while grants and prize competitions could accelerate open benchmarking tools and usability testing methods for real-world clinical workflows [10-12].

Question 5: How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

Accreditation, certification, and industry-driven testing can lower barriers to adoption and improve safety if they evaluate both the AI product and the deploying organization, using a tiered approach matched to clinical risk. A practical structure is a layered governance-to-certification model (regulation and policy aligned to standards, standardized assessment procedures, and reproducible testing tools/metrics), with two complementary credentials: an “AI product” certification (intended use, external validation, robustness and bias testing, transparency documentation) and a “responsible deployment” certification for health systems and clinics (governance, clinician training, monitoring dashboards, and safety-event reporting) [13, 14]. This approach aligns with the National Academy of Medicine’s AI Code of Conduct framework, which emphasizes the value of a shared code of conduct to guide safe integration of AI into clinical practice and strengthen trust among clinicians and patients [15]. This also aligns with the growing view that AI utility depends not only on algorithm performance, but also on health system capacity, workflow integration, and downstream clinical actions; proactive design using target product profiles (TPPs) can help ensure tools are clinically useful and economically sustainable [16].

To make certification actionable rather than “principles-only,” HHS can encourage standardized documentation deliverables (for example, risk-scaled disclosure requirements and structured documentation such as model cards/model documentation, performance by subgroup, limitations, and post-deployment monitoring plans), while balancing transparency with privacy, intellectual property, and security constraints [17]. In addition, transparency documentation should make explicit any embedded value assumptions and tradeoffs (for example, how models balance cost, resource allocation, clinical outcomes, and patient autonomy), as proposed in the “Values in the Model” (VIM) framework, to support trust and responsible governance [18].

HHS can best support these activities by harmonizing expectations across agencies and programs and by incentivizing adoption of credible certification pathways rather than creating duplicative requirements. Specifically, HHS can convene or fund national testing frameworks and shared toolkits for pre-deployment evaluation (dataset provenance and representativeness, subgroup performance reporting, robustness testing) and post-deployment monitoring (drift, calibration, equity metrics), aligned with healthcare-focused responsible use guidance and with confidential learning systems for near-misses and harms (e.g., Patient Safety Organization-style reporting) [11]. HHS can then make certification “real” by linking it to programmatic levers—CMS demonstrations or incentives, procurement expectations for federally supported providers, and grant conditions for AI development and implementation—so high-quality tools are easier to adopt in community settings without increasing clinician burnout or total costs. Endocrine-relevant examples include certification expectations for inpatient insulin safety and hypoglycemia prevention systems and for diabetes technology data integration tools that must work across EHRs and device vendors, with harmonized data elements and analytic definitions, to support equitable access [11, 19].

Question 6: Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

AI has met expectations most reliably in narrow, well-defined tasks that remove manual work or reduce common, avoidable harm, especially when integrated into workflows with clear accountability. Recent U.S. health system experience suggests that ambient documentation (“ambient notes”) is among the most widely adopted use cases and is reported as relatively successful early on, consistent with the idea that tools reducing clerical burden can improve efficiency without needing to “outperform” clinicians on complex diagnostic decisions [20, 21]. In endocrine care, similarly bounded, high-value examples include models that identify hospitalized patients at risk for iatrogenic hypoglycemia (supporting earlier prevention and safer insulin use) and tools that streamline review of continuous glucose monitoring

(CGM) data so clinicians can focus time on counseling and management decisions rather than manual data reconciliation [11, 22].

AI has fallen short when tools are deployed without strong external validation, when they generate excessive alerts, or when they shift work to clinicians without improving outcomes; real-world experience also highlights "immature" tools, financial concerns, and regulatory uncertainty as common barriers to sustained success [11, 20]. Some widely deployed clinical areas—such as imaging/radiology and broad risk stratification (e.g., early sepsis detection)—have shown mixed success across organizations, reinforcing the need for rigorous evaluation, calibration, and post-deployment monitoring rather than assuming performance transfers across sites [20, 21, 23]. The most promising novel tools are those that demonstrably improve outcomes and equity while reducing administrative burden: inpatient glycemic safety support; diabetes technology data integration and personalization; obesity and metabolic disease access/navigation support; and thyroid nodule risk stratification and follow-up coordination that reduces duplicative imaging and missed surveillance. These priorities require ongoing monitoring for drift and subgroup performance to avoid widening disparities and to prevent downstream spending from increased testing or referrals without benefit [10, 11, 21]. Cost-shifting is a real concern; AI implementations should be evaluated on total cost of care and downstream utilization, not just local operational savings.

Question 7: Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Within health care organizations, adoption is usually driven jointly by clinical leadership (quality and patient safety), informatics/IT leadership (EHR integration and data governance), compliance/privacy and security officers, and financial leadership (value and sustainability). In practice, tools that affect medication safety, documentation, or patient communication also require strong frontline engagement from nurses, pharmacists, and clinicians, because usability and workflow fit determine whether AI reduces burden or creates alert fatigue and burnout [5, 11, 20]. In endocrinology, governance often must span inpatient and outpatient teams and incorporate diabetes device data (continuous glucose monitoring and insulin delivery), which increases implementation complexity but is essential for safety and outcomes [11, 24].

Primary administrative hurdles include procurement and contracting (including data rights, security requirements, and indemnification), EHR integration and interoperability constraints, privacy/security review, clinician training and change management, establishing monitoring dashboards and incident response processes, and demonstrating value without increasing costs. Real-world experience shows that even when AI performs well in studies, organizations often struggle to translate that success into routine practice without shared implementation playbooks, dedicated governance, and ongoing monitoring [25, 26]. HHS can help by supporting standardized implementation toolkits and evaluation methods, reducing duplicative reviews through clearer national expectations, and promoting interoperability so endocrine AI tools can be adopted more broadly across community and rural settings—improving access while maintaining safety and equity [5, 11, 20].

Question 8: Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Enhanced interoperability is a patient-safety issue. When endocrine patients move between outpatient and inpatient care (or between health systems), incomplete or delayed data sharing increases the risk of hypoglycemia/hyperglycemia, medication errors, duplicated testing, and missed follow-up. Enhanced interoperability also supports patient-centered care by enabling patients to share timely updates in symptoms, preferences, and patient-reported outcomes across care settings. This can improve patient experience and allow recommendations and care plans to reflect changes in clinical status rather than relying on outdated snapshots. Improving interoperability reduces the need for custom, one-off interfaces and makes it more realistic to deploy clinically useful AI broadly—not just in a few well-resourced sites—while supporting equitable access for rural, lower socioeconomic, and underrepresented populations.

In endocrinology, the highest-yield data to make reliably shareable are: diabetes technology streams (continuous glucose monitoring time-series data, insulin pump/pen dosing, automated insulin delivery settings, hypoglycemia events and alerts), core medications and labs that drive decisions (insulin, GLP-1/GIP agents, glucocorticoids; A1c/glucose/ketones; kidney function; lipids; thyroid tests), thyroid ultrasound plus pathology (imaging objects and structured reports; FNA cytology; surgical pathology and molecular testing), and relevant patient-reported outcomes (hypoglycemia burden, symptoms, side effects, quality-of-life measures). Interoperability for these data types supports AI that can reduce preventable harm (e.g., inpatient insulin safety), improve outcomes (e.g., earlier identification of hypoglycemia risk), and reduce clinician burden by automating data gathering and reconciliation rather than adding new documentation tasks.

A practical standards path already exists and should be reinforced through federal policy, certification, and procurement expectations: FHIR-based exchange using the US Core Implementation Guide to define a consistent “minimum set” of data elements and interactions, aligned with the United States Core Data for Interoperability (USCDI) so the same core clinical data are available across certified systems. Consistent coding for observations and labs (e.g., LOINC) is essential so AI tools can be validated and safely reused across sites. Interoperability should also support benchmarking and monitoring so tools improve care without increasing burnout or costs: (1) external validation on multi-site datasets that include community settings and underrepresented groups; (2) ongoing post-deployment monitoring for calibration, drift, and subgroup performance with clear thresholds for action; and (3) governance and incident reporting processes when AI contributes to near-misses or harm—approaches consistent with national risk-management and healthcare-focused responsible-use guidance [3, 11].

Question 9: What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

Patients’ needs and their concerns should be the center of any discussion of changes in their health care delivery.

Key Challenges Patients and Caregivers Want AI to Address
1. Care coordination and unmet social needs
2. Access to timely, personalized, and efficient care
3. Improved communication and responsiveness to patient concerns
4. Reduced administrative burden and improved care experience

Key Patient and Caregiver Concerns About AI Adoption

1. Lack of transparency and insufficient information
2. Loss of human interaction and clinician autonomy
3. Privacy, data security, and misuse of personal health information
4. Risk of bias, inequity, and incorrect decisions
5. Lack of clear accountability and consent processes

Patients and caregivers also have consistent concerns that should be addressed up front: privacy and secondary use of sensitive health data; cybersecurity; lack of transparency about when AI is used; and the risk of biased performance that could worsen disparities for racial/ethnic minorities, people with limited English proficiency, disability, rural residence, or lower socioeconomic status [2, 11, 27, 28]. Patients want the ability to opt out when feasible, and to know when AI is being used. The "digital divide" is particularly relevant in endocrinology because many benefits depend on smartphones, sensors, connectivity, and health literacy; if AI-enabled tools are deployed without deliberate support, they can widen gaps in outcomes and access [28, 29]. Patients and caregivers are also concerned about accountability when AI outputs are wrong, and about "financial toxicity" if AI-driven recommendations lead to more testing or higher-cost care without clear benefit [30]. Practical safeguards include plain-language disclosure and education for patients and caregivers about when AI is used and what it does; clinician-in-the-loop oversight for higher-risk decisions; routine monitoring for safety and subgroup performance; and ensuring that access supports (training, language accessibility, device/connectivity options) are part of implementation, not an afterthought [11, 28, 29].

From a clinical endocrinology perspective, patients and caregivers are most likely to value AI that measurably improves safety, reduces daily self-management burden, and improves access—without adding cost or complexity. In diabetes care, that includes tools that help prevent severe hypoglycemia and dangerous hyperglycemia, simplify review of continuous glucose monitoring (CGM) and insulin dosing data, support safer inpatient insulin use, and help patients get timely education and follow-up (including triage of portal messages and appointment navigation when access is limited). In obesity and metabolic disease, many patients struggle with medication access and administrative barriers (coverage changes, prior authorization, step therapy); appropriately designed tools could reduce delays and errors by supporting standardized documentation and routing patients to the right resources. In thyroid disease, patients want faster, clearer evaluation of nodules and cancer surveillance, with fewer duplicative tests and better coordination across imaging, pathology, and follow-up. Patient-perspective literature emphasizes that acceptance depends on transparency, understandable communication, and clear clinician oversight—patients generally do not want AI to replace the clinician's role in decisions that affect diagnosis or treatment [2, 11, 27].

Question 10: Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

HHS should prioritize applied, patient-safety–focused research that proves real-world benefit across diverse U.S. care settings (academic, community, and rural) without increasing total costs or clinician workload. The following are six high-priority areas to accelerate safe and effective clinical AI adoption:

1. Real-world evaluation and clinical effectiveness research
2. National interoperable data infrastructure and validation networks
3. Implementation science and workflow integration
4. Trustworthy AI: bias mitigation, safety, and transparency

5. Workforce training and human–AI collaboration
6. Primary care and preventive AI applications

These investments will create the scientific, technical, and operational foundation necessary for scalable, safe, and equitable AI integration into clinical care.

In endocrinology, high-yield priorities include: (1) inpatient glycemic safety tools that predict and prevent hypoglycemia and severe hyperglycemia using electronic health record and medication administration data, paired with implementation studies that measure safety events, nurse/physician workload, and alert fatigue; (2) standards-based integration of diabetes technology streams (continuous glucose monitoring and insulin delivery) into clinical workflows, with evaluation of whether decision support improves outcomes and reduces documentation burden; (3) thyroid ultrasound/biopsy/pathology workflow tools that improve risk stratification and follow-up coordination, with external validation across equipment vendors and practice types; (4) equitable obesity/metabolic disease decision support that reduces administrative friction (coverage and prior authorization workflows) while measuring access, time-to-therapy, and downstream utilization; and (5) lifecycle monitoring science (calibration, drift, subgroup performance, and incident reporting) so deployed tools remain safe and fair over time [3, 6, 11].

10a. Are there published findings about the impact of adopted AI tools and their use in clinical care?

Yes, but the strongest evidence tends to be in narrower, well-defined use cases where outcomes and workflows can be measured.

Strong evidence exists for:
- Improved diagnostic accuracy
- Improved workflow efficiency and documentation
- Earlier detection of clinical deterioration
- Improved operational performance

Moderate or mixed evidence exists for:
- Improved patient outcomes
- Improved clinician decision-making in real-world settings

Published evidence demonstrates that AI tools can improve diagnostic accuracy, clinical efficiency, and operational performance in healthcare. AI systems have achieved expert-level performance in imaging tasks such as breast cancer detection, lung nodule classification, and diabetic retinopathy screening, improving diagnostic precision and reducing false positives. Predictive AI models using electronic health record data have enabled earlier identification of clinical deterioration, sepsis risk, and cardiovascular disease, supporting earlier intervention and preventive care. AI tools have also improved clinical workflow efficiency. For example, AI stroke detection systems have reduced time to treatment by rapidly identifying large vessel occlusions, and AI-assisted documentation tools have significantly reduced clinician administrative burden. However, real-world clinical outcome improvements remain variable and depend heavily on proper validation, workflow integration, and human oversight. AI is most effective when augmenting clinician decision-making rather than replacing clinical judgment, underscoring the importance of careful implementation and ongoing monitoring.

In diabetes, multiple studies show that machine learning models can identify hospitalized patients at risk for inpatient (iatrogenic) hypoglycemia, supporting a pathway to prevent harm when coupled to appropriate clinical oversight and workflow design [22, 31]. In thyroid care, independent evaluations of ultrasound decision support tools suggest clinically meaningful diagnostic performance, while also reinforcing the need for multi-site validation, clear labeling of intended use, and monitoring when moved into routine practice [32-34]. More broadly, authoritative guidance emphasizes that "impact" should include patient outcomes and safety events as well as clinician workload and equity, and that post-deployment monitoring is essential for safety and trust [3, 11].

10b. How does the literature approach the cost, benefits, and transfers of using AI as part of clinical care?

The literature increasingly treats cost and transferability as implementation problems, not only model-performance problems. Economic evaluations account for integration (interfaces, training, workflow redesign), ongoing maintenance (updates and cybersecurity), and monitoring (performance drift, recalibration, subgroup auditing), alongside downstream effects on utilization (testing, referrals, avoidable admissions) and clinician time [25, 35].

Published economic evaluations indicate that many AI applications in healthcare are cost-effective or cost-saving, particularly when used to improve diagnostic accuracy, enable earlier detection of disease, and reduce avoidable complications and unnecessary procedures. AI tools that enhance screening, risk prediction, and workflow efficiency can reduce downstream costs by preventing hospitalizations, complications, and redundant testing. However, most studies emphasize that economic benefits depend heavily on successful integration into clinical workflows and sustained real-world performance. Implementation costs—including infrastructure, validation, training, and ongoing monitoring—can be substantial and are often borne by healthcare providers, while financial savings frequently accrue to payers or the broader healthcare system. This misalignment creates cost-shifting that may discourage provider adoption in the absence of clear reimbursement pathways or shared savings models. Additionally, many published economic analyses incompletely account for full lifecycle costs, including maintenance, governance, and retraining, highlighting the need for prospective, real-world economic evaluations. Policies that align reimbursement with demonstrated clinical and economic value will be essential to accelerate responsible AI adoption while ensuring sustainability for healthcare providers.

Endocrine-adjacent screening provides a useful model: autonomous AI for diabetic retinopathy has been evaluated as cost-effective or cost-saving under plausible assumptions because it can increase screening reach while reducing downstream complications, but the value depends on deployment context and full accounting of implementation costs [36]. Other relevant areas include:

a) Predictive models for diabetes complications: AI enables earlier intervention, reducing hospitalizations and complications such as kidney failure or cardiovascular events.
b) Continuous glucose monitoring and AI-assisted insulin management: Improves glycemic control and reduces hypoglycemia-related emergency care.
c) Population health management tools: AI identifies high-risk patients with diabetes or metabolic disease, enabling targeted preventive care and reducing downstream costs.

Systematic reviews highlight wide variability in assumptions and study quality, underscoring the need for transparent reporting of costs, comparators, and setting so results generalize across health systems and do not unintentionally increase spending without benefit [25, 35].

Conclusion

AI is already being integrated into clinical care, but it is crucial that government, as the public's representative, balance judicious safeguards with policies that encourage responsible public- and private-sector innovation.

The American Association of Clinical Endocrinology (AACE) supports the responsible adoption of AI that improves patient safety, outcomes, and access while strengthening trust, protecting privacy, and supporting clinicians and care teams. In diabetes care, endocrinologists already have experience with AI-enabled technologies such as continuous glucose monitoring and automated insulin delivery, underscoring both the feasibility of responsible implementation and the opportunity to extend these benefits to a wider range of patients and care settings. For AI to deliver meaningful benefit in endocrine care at scale, it must be developed and evaluated using representative data; integrated into workflows in ways that reduce administrative burden; monitored over time for performance, safety, and equity; and deployed so that patients in rural, underserved, and underrepresented communities can access the same benefits as those in well-resourced settings. We appreciate HHS's leadership in convening stakeholders on these issues and welcome continued collaboration to ensure that AI advances translate into better, safer, more equitable clinical care. We respectfully urge HHS to synthesize comments around a small set of cross-cutting priorities—interoperability, lifecycle oversight, equity reporting, privacy/security, and payment alignment—to accelerate safe and equitable adoption.

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